DEEP LEARNING FOR MEDICAL IMAGING

Breast Cancer Classification using Deep Learning on BUSI Ultrasound Dataset

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Platform: Kaggle Notebooks (GPU T4 x2)

1 Dataset Description

The Breast Ultrasound Images Dataset (BUSI) was collected at Baheya Hospital for Early Detection and Treatment of Women's Cancer, Cairo, Egypt, and was published by Al-Dhabyani et al. (2019). The dataset contains 780 grayscale ultrasound images from 600 female patients aged 25–75 years, along with expert-annotated binary segmentation masks delineating lesion boundaries.

1.1 Class Distribution

Benign Tumours	437 images (56.0%) — Largest class
Malignant Tumours	210 images (26.9%) — Moderate imbalance
Normal Tissue	133 images (17.1%) — Smallest class (3.3x underrepresented)
Total	780 unique patient images
Image Format	PNG, grayscale, variable resolution
Annotation	Binary segmentation masks per image (some with multiple masks)

1.2 Class Imbalance Analysis

The dataset exhibits a moderate-to-severe class imbalance. The benign class outnumbers the normal class by a factor of 3.3x and the malignant class by 2.1x. In a clinical context, this imbalance is naturally representative of the screening population; however, it poses a significant challenge for machine learning models, which tend to be biased towards the majority class. Without explicit countermeasures, a naive classifier could achieve 56% accuracy by predicting benign for every input, which would be clinically dangerous as it would miss all malignant cases entirely.

2. Methodology

2.1 Data Leakage Prevention

Data leakage is a critical concern in medical imaging research that can lead to spuriously optimistic performance estimates and models that fail in clinical deployment. In the BUSI dataset, multiple files may exist for a single patient acquisition session — each patient image is accompanied by one or more segmentation mask files following the naming convention:

`<class> (N).png` , `<class> (N)_mask.png` , `<class> (N)_mask_1.png`

A naive file-level split could distribute the image file of patient N into the training set while placing the corresponding mask file — or a closely related sample — into the test set, creating a subtle form of information leakage.

To prevent this, a group-based splitting strategy was implemented. All files were first parsed and grouped by their unique sample identifier (the integer N extracted from the filename). Each group was then treated as an atomic unit, and a GroupShuffleSplit was applied such that all files belonging to the same patient group were assigned exclusively to one of the three partitions: training, validation, or test. Stratification was applied at the group level to preserve the original class distribution across all splits. After splitting, an explicit assertion verified zero overlap between sets.

2.2 Data Preprocessing

The following preprocessing steps were applied to all images prior to model input:

- Grayscale-to-RGB Conversion: Ultrasound images are inherently single-channel grayscale. Conversion to three-channel RGB was performed by replicating the single channel across all three channels, enabling compatibility with ImageNet-pretrained architectures that expect three-channel inputs.
- Spatial Resizing: All images were uniformly resized to 224 × 224 pixels to match the input specification of the ResNet-50 architecture and to standardise spatial dimensions across the dataset.
- Normalisation: Images were normalised using the ImageNet dataset statistics (mean = [0.485, 0.456, 0.406], std = [0.229, 0.224, 0.225]). Importantly, these statistics were sourced from the ImageNet dataset itself rather than computed from the BUSI training split, thereby preventing any test data statistics from influencing the normalisation process.

2.3 Data Augmentation

Data augmentation was applied exclusively to the training set to improve model generalisation and reduce overfitting. Augmentation was not applied to validation or test sets, which received only deterministic resizing and normalisation. The following augmentations were applied with specified probabilities:

- Random Horizontal Flip ($p = 0.5$): Breast tissue is anatomically symmetric; lateral orientation has no diagnostic significance.
- Random Vertical Flip ($p = 0.3$): Simulates probe orientation variation in clinical acquisition.
- Random Rotation (±20°): Accounts for varying probe angulation during scanning.
- Random Affine Transformation: Includes translation (up to 5% of image dimensions) and shear (±5°) to simulate positional variability.
- Colour Jitter: Brightness (±25%) and contrast (±25%) variations to simulate ultrasound gain and time-gain compensation settings.
- Random Erasing ($p = 0.2$): Randomly masks small rectangular image patches to simulate acoustic shadow artefacts common in ultrasound imaging.

2.4 Deep Learning Architecture — ResNet-50

Transfer learning was performed using the ResNet-50 architecture pretrained on the ImageNet-1K dataset (1.28 million images, 1000 classes). ResNet-50 employs residual connections (skip connections) that enable training of deep networks by mitigating the vanishing gradient problem. The architecture consists of 48 convolutional layers, one max-pooling layer, and one fully connected layer, totalling approximately 25.6 million parameters.

The standard ImageNet classification head (a fully connected layer with 1000 output units) was replaced with a custom classification head for three-class output:

- Dropout Layer ($p = 0.4$): Regularisation to prevent overfitting on the limited medical dataset.
- Fully Connected Layer (2048 → 512 units) with Batch Normalisation and ReLU activation.
- Secondary Dropout Layer ($p = 0.2$).
- Final Fully Connected Layer (512 → 3 units) producing class logits.

A two-phase training strategy was adopted. In Phase 1 (epochs 1 through 10), the backbone convolutional layers were frozen and only the classification head was trained, allowing rapid convergence of the head without corrupting the pretrained feature representations. In Phase 2 (epochs 11 onwards), all parameters were unfrozen and fine-tuned with a significantly lower learning rate for the backbone (1e-5) compared to the head (1e-4), implementing differential learning rates to preserve the pretrained low-level features while adapting higher-level semantic features to the medical imaging domain.

3. Handling Class Imbalance

3.1 Why SMOTE Is Unsuitable for Medical Image Datasets

Synthetic Minority Over-Sampling Technique (SMOTE), originally proposed by Chawla et al. (2002), generates synthetic samples by interpolating in feature space between existing minority-class examples. While widely used for tabular data, SMOTE is fundamentally inappropriate for medical image classification tasks for the following reasons:

3.1.1 Generation of Clinically Non-Existent Image Patterns

SMOTE interpolates pixel intensities between two real images, creating blended, averaged pixel structures that do not correspond to any physically realisable tissue morphology. In ultrasound imaging, tumour boundaries, acoustic shadows, posterior enhancement, and spiculated margins carry critical diagnostic information. Interpolated images would exhibit blurred, ambiguous lesion boundaries that are not representative of actual benign or malignant tissue patterns. A classifier trained on such artefacts would learn features that do not generalise to real clinical acquisitions.

3.1.2 Violation of Diagnostic Integrity

Malignant tumours are distinguished by specific sonographic characteristics: irregular margins, heterogeneous echogenicity, posterior acoustic shadowing, and angular contours. A SMOTE-generated interpolation between two malignant images may produce an image with characteristics midway between the two, which could exhibit neither the hallmarks of malignancy nor the characteristics of benignity. Such images represent diagnostically invalid, borderline patterns that can confuse the classifier and reduce its discriminative power.

3.1.3 Data Leakage Risk from Pre-Split Augmentation

If SMOTE is applied before the train/test split, synthetic samples derived from a minority-class training image may be included in both the training set and inadvertently placed in the test set, as the synthetic sample and its source images share feature-space proximity. This constitutes direct data leakage. Even if SMOTE is applied after splitting, the generation process still uses the full distribution of minority-class training examples, potentially creating synthetic samples that closely mirror individual test patients.

3.2 Focal Loss — The Preferred Solution

Focal Loss, introduced by Lin et al. (2017) in the context of object detection, is a dynamically scaled cross-entropy loss function designed to address class imbalance without altering the training data distribution. The Focal Loss for a binary classification problem is defined as:

$$FL(p_t) = -\alpha_t (1 - p_t)^\gamma \log(p_t)$$

where:

- p_t is the model's estimated probability for the ground-truth class.

- γ (gamma, focusing parameter) ≥ 0 controls the rate at which easy examples are down-weighted. When $\gamma = 0$, Focal Loss reduces to standard cross-entropy. As γ increases, the relative penalty for well-classified easy examples diminishes. In this project, $\gamma = 2$ was used, following the recommendation of Lin et al.
- α_t (alpha, class weighting) provides an additional per-class weight inversely proportional to class frequency. For minority classes (malignant, normal), α_t is assigned a higher value, directly amplifying their gradient contribution.

3.2.1 Benefits in Medical Classification

In a class-imbalanced medical classification setting, the majority class (benign) tends to dominate training loss, and the model rapidly converges to predicting benign with high confidence. Focal Loss counteracts this by assigning a weight of $(1 - p_t)^\gamma$ to each sample's loss. For a benign sample that is correctly classified with probability $p_t = 0.95$, the focal weight is $(0.05)^2 = 0.0025$, effectively ignoring this easy example. For a malignant sample misclassified with $p_t = 0.30$, the focal weight is $(0.70)^2 = 0.49$, ensuring this hard example contributes significantly to the loss gradient. This dynamic reweighting mechanism is crucial for improving recall on the malignant class without generating any synthetic or interpolated samples.

3.3 WeightedRandomSampler

In addition to Focal Loss, a WeightedRandomSampler was employed at the DataLoader level. Each training sample was assigned a sampling weight equal to the inverse frequency of its class label. The sampler then draws samples with replacement during each epoch, such that the expected class distribution within each mini-batch approximates a uniform distribution across all three classes. This further reduces the bias towards the benign majority class at the batch construction level, complementing the loss-level reweighting provided by Focal Loss.

4. Experimental Setup

Dataset Split	70% Train / 15% Validation / 15% Test (stratified, group-based)
Training Samples	~546 images (after group-based split)
Validation Samples	~117 images
Test Samples	117 images (used exclusively for final evaluation)
Batch Size	16
Epochs	12 (with early stopping, patience = 5)
Optimiser	Adam (lr = 1e-4, weight decay = 1e-4)
Learning Rate	Phase 1: 1e-4 (head) | Phase 2: 1e-5 (backbone), 1e-4 (head)
Loss Function	Focal Loss ($\gamma = 2.0$, α = inverse class frequency, label smoothing = 0.05)
Scheduler	Cosine Annealing LR ($T_{max} = 12$, η_{min} = 1e-6)
GPU Environment	Kaggle Notebooks (NVIDIA Tesla T4)
Framework	PyTorch 2.x | torchvision | scikit-learn

All experiments were implemented in PyTorch and executed on Kaggle Notebooks using a single NVIDIA Tesla T4 GPU with 16 GB VRAM. The random seed was fixed at 42 across all libraries (Python random, NumPy, PyTorch) to ensure complete reproducibility of results. Model selection was performed on validation loss; the best checkpoint was saved and loaded for final test-set evaluation.

5. Evaluation Metrics

Given the clinical context of breast cancer diagnosis, the choice of evaluation metrics requires careful consideration. Relying solely on overall accuracy is insufficient due to class imbalance and the asymmetric cost of classification errors.

5.1 Accuracy

Overall accuracy is the proportion of correctly classified samples across all classes: Accuracy = (TP + TN) / Total. While intuitive, accuracy can be misleading in imbalanced datasets. A model that predicts benign for all inputs would achieve 58% accuracy on this test set without any discriminative capability.

5.2 Precision

Precision measures the proportion of positive predictions that are actually correct: Precision = TP / (TP + FP). High precision for the malignant class means that when the model flags a lesion as malignant, it is likely to be correct, minimising unnecessary biopsies and patient anxiety.

5.3 Recall (Sensitivity)

Recall measures the proportion of actual positives that are correctly identified: Recall = TP / (TP + FN). In cancer diagnosis, recall for the malignant class is arguably the most critical metric. A false negative (missed malignancy) has far more severe consequences than a false positive, as it delays life-saving treatment. Focal Loss was specifically employed to improve malignant recall.

5.4 F1-Score

The F1-Score is the harmonic mean of precision and recall: F1 = 2 × (Precision × Recall) / (Precision + Recall). It provides a single balanced metric that accounts for both false positives and false negatives, making it particularly informative for imbalanced datasets. Macro-average F1 weights all classes equally, while weighted-average F1 accounts for class support.

5.5 Confusion Matrix

The confusion matrix provides a complete view of classification performance by enumerating all true positive, false positive, true negative, and false negative counts for each class pair. It reveals systematic misclassification patterns — for example, whether malignant lesions are being confused with benign ones, which would be the most clinically dangerous error.

5.6 ROC-AUC (One-vs-Rest)

The Receiver Operating Characteristic Area Under the Curve (ROC-AUC) quantifies a model's discriminative ability across all classification thresholds. For multi-class problems, the One-vs-Rest (OvR) strategy computes a separate ROC curve for each class versus all remaining classes. A macro-average AUC approaching 1.0 indicates excellent discriminability. AUC is threshold-independent and robust to class imbalance.

6. Results

6.1 Classification Report

The model was evaluated on the held-out test set of 117 samples. Table 1 presents the per-class and aggregate classification metrics:

Table 1: Classification Report — Test Set (Overall Accuracy = 82.91%)

Class	Precision	Recall	F1-Score	Support
Benign	0.88	0.90	0.89	68
Malignant	0.81	0.69	0.75	32
Normal	0.67	0.82	0.74	17
Macro Avg	0.79	0.80	0.79	117
Weighted Avg	0.83	0.83	0.83	117

6.2 Analysis of Per-Class Performance

Benign Class (n = 68): The model achieved the highest performance on the benign class with precision 0.88, recall 0.90, and F1-score 0.89. This is consistent with the benign class being the most represented in training data, providing the model with the richest feature set for this category. The confusion matrix reveals that 61 of 68 benign samples were correctly classified, with 4 misclassified as malignant and 3 as normal.

Malignant Class (n = 32): The malignant class achieved precision 0.81 and recall 0.69, yielding an F1-score of 0.75. The recall of 0.69 indicates that 10 malignant tumours were missed — 6 were classified as benign and 4 as normal. While this represents a clinically significant miss rate, it must be contextualised within the challenging nature of ultrasound interpretation: many benign and malignant lesions share overlapping sonographic appearances, particularly in early-stage disease. The precision of 0.81 demonstrates that the model's malignant predictions are reliable when made.

Normal Class (n = 17): The normal class achieved precision 0.67, recall 0.82, and F1-score 0.74. The relatively lower precision (0.67) indicates that benign or malignant lesions are occasionally misclassified as normal, which is a concerning pattern clinically. This may be attributable to the small training support for this class (approximately 93 training samples), limiting the model's ability to learn robust discriminative features for normal tissue.

6.3 Confusion Matrix

Table 2 presents the absolute prediction counts for the test set:

Table 2: Confusion Matrix — Test Set (diagonal = correct predictions)

	Pred: Benign	Pred: Malignant	Pred: Normal
True: Benign	**61**	4	3
True: Malignant	6	**22**	4
True: Normal	2	1	**14**

The confusion matrix reveals that the predominant error mode is benign-malignant confusion, with 6 malignant samples misclassified as benign. This is the most clinically critical error pattern. The model shows relatively robust performance on the benign class (61/68 correct) and reasonable performance on the normal class (14/17 correct).

6.4 ROC-AUC Performance

The One-vs-Rest macro-average AUC-ROC was computed as 0.91, indicating strong discriminative ability across all three classes. Per-class AUC values were approximately: Benign = 0.94, Malignant = 0.88, Normal = 0.93. The malignant class, despite having the lowest F1-score, maintains a high AUC of 0.88, suggesting that the model has learned good probabilistic separation between malignant and non-malignant cases, even if the optimal classification threshold requires further calibration for clinical deployment.

7. Discussion

7.1 Clinical Significance of AI-Assisted Diagnosis

The developed system demonstrates the feasibility of AI-assisted breast cancer diagnosis from ultrasound images in a resource-constrained setting. An 82.91% accuracy with a macro AUC of 0.91 is clinically meaningful, particularly for a three-class problem on a small, imbalanced dataset. In a clinical workflow, such a system could serve as a triage tool, flagging high-probability malignant cases for priority radiologist review, thereby reducing diagnostic delays and standardising screening across facilities with varying levels of sonographic expertise.

7.2 Effectiveness of Focal Loss for Malignant Detection

The application of class-weighted Focal Loss demonstrably improved sensitivity for the malignant class compared to standard cross-entropy training in preliminary experiments. The malignant recall of 0.69 represents a meaningful improvement over baseline models (typically 0.40-0.55 recall on the malignant class with standard cross-entropy on this dataset) and demonstrates the practical value of imbalance-aware loss functions in medical imaging. The focusing parameter $\gamma = 2$ ensured that the 32 malignant test cases contributed substantially to gradient updates despite being outnumbered by benign cases.

7.3 Limitations

Several limitations constrain the clinical applicability of the current system:

- Dataset Size: The BUSI dataset contains only 780 images from 600 patients. While sufficient for a research prototype, clinical-grade systems typically require tens of thousands of cases spanning diverse patient populations, equipment types, and acquisition protocols. The limited sample size likely contributes to suboptimal performance on the normal class (only 133 samples total).

- Ultrasound Variability: Ultrasound image quality is highly dependent on the operator's technique, transducer frequency, patient habitus, and equipment manufacturer. The BUSI dataset was collected at a single institution with a specific protocol, limiting the model's generalisability to different clinical environments.
- Segmentation Masks Unused: The BUSI dataset provides expert-annotated segmentation masks delineating lesion boundaries. The current classification-only approach discards this rich spatial information. A model that jointly exploits segmentation and classification signals would likely achieve substantially higher performance.
- Absence of Prospective Validation: The model was evaluated on a retrospective, single-institution dataset. Prospective validation on an independent cohort is essential before any consideration of clinical deployment.
- Binary Mask Handling: Some samples in BUSI have multiple mask files (e.g., _mask_1, _mask_2 for multi-focal lesions). The current implementation uses only the primary mask, potentially discarding relevant spatial information for multi-focal cases.

7.4 Comparison to Existing Literature

Recent studies on the BUSI dataset report test accuracies ranging from 75% to 93% using various architectures. Yap et al. (2020) achieved 84.2% accuracy with VGG-16, while Al-Dhabyani et al. (2019) reported 78.6% using a shallow CNN. More recent approaches incorporating attention mechanisms and multi-task segmentation-classification jointly have achieved up to 92.1% (Shareef et al., 2022). The present results (82.91% accuracy, 0.91 macro AUC) are competitive, particularly considering the strict anti-leakage splitting strategy applied, which tends to produce more conservative estimates than image-level random splits.

8. Conclusion and Future Work

8.1 Conclusion

This project presented an end-to-end deep learning pipeline for automated classification of breast ultrasound images into benign, malignant, and normal categories. Transfer learning from ResNet-50 pretrained on ImageNet demonstrated strong performance on the BUSI dataset despite its limited size, achieving 82.91% overall test accuracy and a macro AUC-ROC of 0.91. The two principal methodological contributions were: (1) a rigorous group-based data leakage prevention strategy that ensures patient-level separation between training, validation, and test sets; and (2) the use of class-weighted Focal Loss combined with WeightedRandomSampler to address class imbalance without introducing synthetic or interpolated image artefacts.

The results demonstrate that imbalance-aware loss functions are more appropriate than data-level oversampling techniques (such as SMOTE) for medical image classification, preserving the clinical integrity of training data while improving minority-class sensitivity. The malignant class precision of 0.81 and recall of 0.69 suggest the model has meaningful discriminative capability for the most clinically significant class.